

Assignment 5: Linear programming

Integrated bioenergy supply chain and interplant heat integration

The multifaceted problem associated with the use of fossil fuels for energy generation has brought about the need for not only alternative cleaner energy sources, but also efficient design of energy networks as well as conservation of resources. The construction of eco-industrial parks, where industrial plants are co-located, for the purpose of resource sharing, is becoming common. Beyond co-locating plants within eco-industrial parks, there is also the need to integrate the resource flow of agricultural biomass, including wastes, as well as other forms of energy rich sources, with industrial and domestic heat demand.

In this assignment, you are required to integrate a set of available bio-based feedstocks, from which energy can be generated, with the hot utility demand of a set of process plants co-located few kilometers from the biobased feedstocks. Shown in Figure 1 is a superstructure representation of the proposed integrated network.

Use the data presented in Tables 1 to 7 to do the following:

1. By setting up the energy demand profile of each plant as a linear programming model, determine the minimum hot and cold utility demand for plants 1, 2 and 3 using the following minimum approach temperatures, Plant 1: $\Delta T_{min} = 10\text{ }^{\circ}\text{C}$, Plant 2: $\Delta T_{min} = 20\text{ }^{\circ}\text{C}$, Plant 3: $\Delta T_{min} = 30\text{ }^{\circ}\text{C}$. Note that you must list the optimum values obtained for each variable in Plants 1, 2 and 3. You may use any of Excel solver, GAMS, Scilab or any other solver environment for this section.
(24 marks, i.e., 8 marks for each plant)
2. Set up the bioenergy supply chain model as a linear program using the data provided in Tables 1 to 7.
3. Integrate the bioenergy supply chain model with the hot utility demand of plants 1, 2 and 3. This can be done by setting the minimum hot utility obtained for each of plants 1, 2 and 3 in section 1 as demand nodes (specified as parameters) in your bioenergy supply chain model. Note that this must also be done for the cold utilities. However, for the hot utilities, their unit cost is determined by the quantity of biomass energy feedstock selected, including the transport mode, while for the cold utilities, the direct unit cost is given in Table 4.
4. Determine the minimum total annual cost (in \$/y) of the integrated network. Also, using a network diagram show the optimum values obtained for all the variables in your model. (30 marks)
5. Present a detailed discussion of your results. (16 marks)

Note that since this is a simplified representation of the network, you must state your assumptions. For the pipeline mode of transport, you may assume that biogas is generated from the feedstock at the supply locations. The biogas is then transported to the process plants where the hot utility (steam) is then generated. For the other two modes of transport, you may assume that the feedstock is converted to biogas at the supply location or at the respective process plants.

To help you develop the bioenergy supply chain network model and maybe give you a clue as to how to integrate the bioenergy supply chain model with the hot utility demand of plants 1, 2 and 3, see Cowen et al. (2019) and Isafiade et al. (2022).

Table 1: Feedstock types, cost, and capacity (modified after Cowen et al. 2019)

		LHV_m (kWh/kg)	Cost (\$/kg)	Maximum supply capacity (kg/y) $\times 10^6$
Supply 1	Corn stover	4.63	0.025	0.50
Supply 2	Glycerol	4.75	0.050	0.45
Supply 3	Wood	4.28	0.060	24.0

In Table 1 LHV_m is lower heat value.

Table 2: Average distances (in km) between biomass supply locations and process plants for various transport modes

Supply	Process plant	Truck	Railway	Pipeline
Supply 1	Plant 1	12	17	14
	Plant 2	22	27	10
	Plant 3	14	15	13
Supply 2	Plant 1	16	20	18
	Plant 2	18	11	14
	Plant 3	10	14	12
Supply 3	Plant 1	7	14	15
	Plant 2	12	9	18
	Plant 3	14	13	8

Table 3: Transport costs

Transport mode	Cost $\$/ (kg \cdot km)$
Truck	0.008
Railway	0.006
Pipeline	0.009

Table 4: Other data

	Data
Boiler efficiency	80%
Operating hours	8000 hours
Cost of cold utility	6 $\$/ (kW \cdot y)$

Table 5: Stream data for Plant 1

Process streams		$T^s(^{\circ}C)$	$T^t(^{\circ}C)$	$FCP(kW/^{\circ}C)$
Hot streams	H1	360	40	18
	H2	210	50	20
	H3	340	210	22
	H4	390	100	16
Cold streams	C1	110	410	20
	C2	50	320	12
	C3	60	400	14
	C4	40	310	10
	C5	60	440	15

Table 6: Stream data for Plant 2

Process streams		$T^s(^{\circ}C)$	$T^t(^{\circ}C)$	$FCP(kW/^{\circ}C)$
Hot streams	H1	350	50	26
	H2	240	60	25
	H3	360	200	20
	H4	410	110	18
Cold streams	C1	120	440	17
	C2	50	310	20
	C3	40	410	21
	C4	45	310	17

Table 7: Stream data for Plant 3

Process streams		$T^s(^{\circ}\text{C})$	$T^t(^{\circ}\text{C})$	$FCP(kW/^{\circ}\text{C})$
Hot streams	$H1$	400	90	8
	$H2$	290	80	6
	$H3$	340	210	14
	$H4$	420	80	9
Cold streams	$C1$	130	220	12
	$C2$	60	200	14
	$C3$	70	320	9
	$C4$	50	480	13

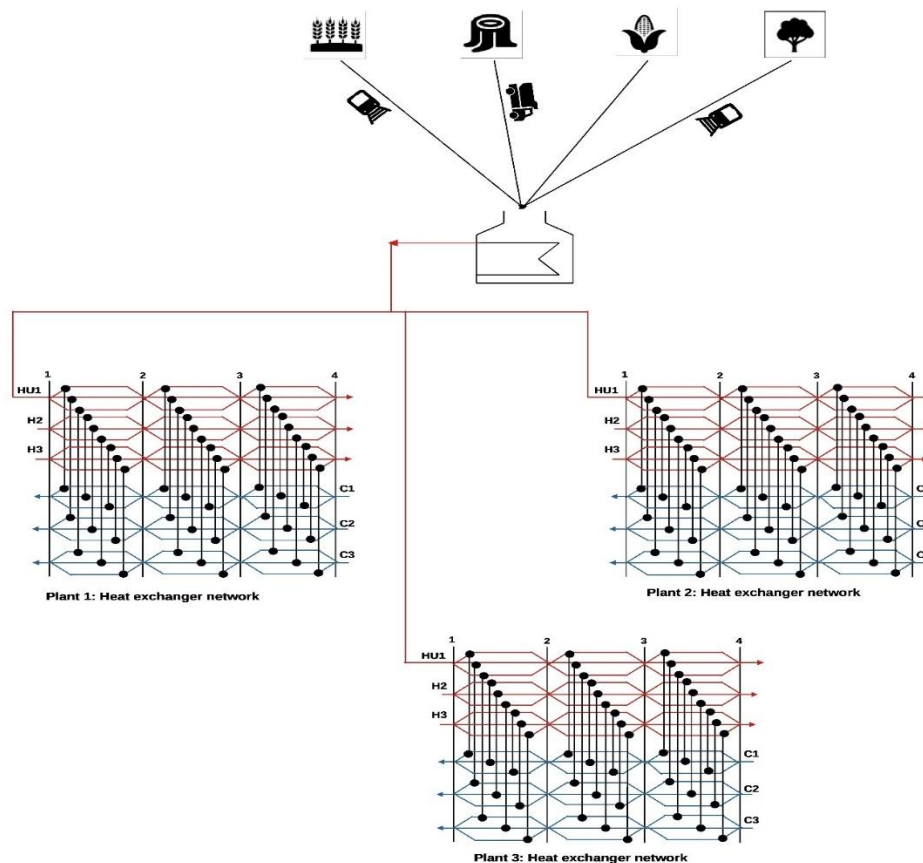


Figure 1: Superstructure representation of integrated bioenergy supply chain and interplant heat exchange network synthesis

References

- Cowen N., Vogel A., Isafiade A.J., Cucek L., Kravanja Z., (2019), Synthesis of Combined Heat Exchange Network and Utility Supply Chain, Chemical Engineering Transactions, 76, 391-396.
<https://doi.org/10.3303/CET1976066>
- Isafiade A.J., Cowen N., Vogel A., Cucek L., Kravanja Z., (2022), Synthesis of Utility Supply Chain Network and Industrial Symbioses for Heat Integration. Journal of Cleaner Production. 380(2022) 134921.
<https://doi.org/10.1016/j.jclepro.2022.134921>